

Minh Tuan Pham

Ho Chi Minh • phamminhtuan1999@gmail.com • +84 941 *** **

Professional Summary

Senior Backend Engineer with over 4 years of experience architecting high-availability distributed systems for government, healthcare, and e-commerce. Specialist in **Node.js (NestJS)** and **.NET 8**, with a proven track record of **slashing API latency by 75%** through strategic caching and query optimization. Expert in **AWS** cloud infrastructure and **event-driven architectures**. Committed to **Test-Driven Development (TDD)** and clean code principles to ensure 99.9% system stability in high-volume environments.

Key Skills

Languages/ Frameworks: Node.js (NestJS), ASP.NET Core (.NET 8), TypeScript, Python.

Backend & Databases: SQL Server, PostgreSQL, MongoDB, Redis, ElasticSearch.

Frontend: ReactJS, Razor Pages, HTML, CSS.

Architecture & DevOps: Microservices, Event-Driven Systems, Docker, CI/CD, Git, Jira.

Cloud Platforms: AWS (EC2, RDS, S3, CloudWatch, SNS), Azure.

Development Practices: Agile (Scrum), Code Review, SOLID Principles.

Work Experience

DATAHOUSE ASIA

Senior Software Engineer

Oct 2022 - Present

- **Architected** core Web APIs for the **Government Disclosure System**, led the backend development of a high-security financial filing system, integrating **stateless authorization** with **complex audit trails** to ensure compliance for state-level financial transactions.
- **Engineered** high-performance **NestJS microservices** for a multi-tenant healthcare ecosystem, reducing response times from 6s to 1.5s via Redis caching and PostgreSQL optimization.
- **Managed** multi-tenant deployments on **AWS**, maintaining **99.9% uptime** through automated scaling and proactive monitoring.
- Led technical delivery and code quality through **backlog refinement**, **pull request reviews**, and promoted **TDD** patterns to minimize production regressions.

MOBILE WORLD GROUP

.NET Developer

Oct 2020 – Sep 2022

- **Engineered** backend workflows for **Tier-1 Retail Website**, supporting an e-commerce catalogue of (12,000+ SKUs), optimizing .NET Core services to maintain system stability during peak-traffic surges.
- Integrated **ElasticSearch** and **Redis** for distributed caching and rapid search indexing, significantly offloading primary SQL Server databases during peak traffic.
- **Refactored** backend transaction modules to ensure strict data consistency between product inventory and checkout services.
- **Optimized** server-side response logic and content delivery structures, resulting in first-page Google rankings and increased organic traffic.

Projects

Government Disclosure System

Client: US Government | 10 team members

Technologies: ASP.NET Core (.NET 8), C#, Razor Pages, Entity Framework Core, SQL Server

- Built end-to-end capabilities for the Financial Transactions Team: APIs and Portal to create, validate, and submit receipts, invoices, and transactions.
- Implemented compliance-ready validation rules, audit logging, and secure access patterns to improve data integrity and support downstream auditing and public search use cases.

- Designed SQL Server data access and managed EF Core migrations to keep schema changes reliable across local and product environments.
- Owned delivery scope and code quality (backlog refinement, pull request reviews) and integrated a background job queue for asynchronous post-filing processing.

Healthcare Management System

Client: Philippines Government | 30 team members

Jun 2023 – Present

Technologies: ReactJS, NestJS, PostgreSQL, Redis, AWS

- Developed backend services using NestJS in a microservice architecture to support scalable, modular deployment for hospital and clinic systems.
- Implemented event-driven services with PostgreSQL and Redis to ensure real-time, fault-tolerant data synchronization across multi-tenant clients.
- Utilized AWS cloud infrastructure end-to-end (EC2, RDS, S3, CloudWatch, SNS, etc.) to deploy, monitor, and scale the platform with 99.9% uptime.

IPTS - Tax Management System

Client U.S. Government | 15 team members

Oct 2022 – May 2023

Technologies: Angular 13, .NET Core, SQL Server, Azure, Event-driven microservices

- Focused on bug resolution and backend enhancements during the production release phase, ensuring system stability and performance for US tax users.
- Developed backend features to support calculations for multiple tax billing types, handling complex financial logic with .NET Core and SQL Server.

Education & Certifications

- **B.S. Software Technology** | Ho Chi Minh University of Technology (Scholarship Recipient).
- **AWS Certified Solutions Architect – Associate** (Mar 2025).
- **IELTS 6.5** (Professional English Proficiency).